Experiment 3

Dimensionality Reduction Using Principal Component Analysis (PCA)

Subject: Data Science Lab | Dataset: Titanic | Mapped CO(s): CO2, CO4

1. Aim

To perform dimensionality reduction using PCA on the Titanic dataset.

2. Key Concepts & Definitions

Dimensionality Reduction

Reducing number of features while retaining important information.

PCA

Transforms correlated variables into uncorrelated components preserving maximum variance.

Variance

Measure of data spread.

$$\text{Var}(X) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

Covariance Cov(X,Y)

How two variables change together.

$$\text{Cov}(X,Y) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

Covariance Matrix C

Matrix of pairwise covariances - input to PCA.

$$\mathbf{C} = \frac{1}{n-1}\mathbf{X}^T\mathbf{X}$$

Standardization (Z-score)

Scale to mean 0, std 1.

$$z = \frac{x-\mu}{\sigma}$$

Eigenvalue (lambda)

Variance captured by each principal component.

Eigenvector (v)

Direction of a principal component.

$$\mathbf{Cv} = \lambda\mathbf{v}$$

Principal Component

New uncorrelated variable.

$$PC_k = \mathbf{X}\mathbf{v}_k$$

Loadings

Feature weights (eigenvector components) showing PC composition.

Explained Variance Ratio

Fraction of total variance per PC.

$$\text{EVR}_k = \frac{\lambda_k}{\sum_{i=1}^{p} \lambda_i}$$

Cumulative Explained Variance

Total variance retained by first k PCs.

$$\text{CEV}_k = \sum_{i=1}^{k} \text{EVR}_i$$

Orthogonality

PCs are uncorrelated.

$$PC_i \cdot PC_j = 0 \quad \text{for } i \neq j$$

3. Symbol Glossary

Symbol	Meaning
X	Standardized data matrix (n × p)
n	Number of samples (714)
p	Number of features (5)
mu	Population/sample mean
sigma	Standard deviation
C	Covariance matrix
lambda	Eigenvalue
v	Eigenvector
W	Matrix of eigenvectors
Z	Transformed PCA data

4. Important Formulae (Summary)

$$z = \frac{x - \mu}{\sigma}$$

$$\text{Cov}(X, Y) = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$$

$$\mathbf{C} = \frac{1}{n-1} \mathbf{X}^T \mathbf{X}$$

$$\mathbf{C}\mathbf{v} = \lambda \mathbf{v}$$

$$\mathbf{Z} = \mathbf{X}\mathbf{W}$$

$$PC_k = \mathbf{X}\mathbf{v}_k$$

$$\text{EVR}_k = \frac{\lambda_k}{\sum_{i=1}^{p} \lambda_i}$$

$$\text{CEV}_k = \sum_{i=1}^{k} \text{EVR}_i$$

$$PC_i \cdot PC_j = 0 \quad \text{for } i \neq j$$

5. Algorithm

- 1. Select numerical features; remove missing values

- 2. Standardize with StandardScaler
- 3. Compute covariance matrix; find eigenvalues/eigenvectors
- 4. Transform data; compute explained variance; plot PC1 vs PC2

6. Experimental Results

- 714 rows, 5 features -> 2 PCs (67.10% variance retained)

PC	Variance %	Cumulative %
PC1	34.83	34.83
PC2	32.27	67.10
PC3	13.82	80.92

Feature Loadings

Feature	PC1	PC2
age	-0.5459	-0.1972
fare	-0.4201	0.5267
sibsp	0.3115	0.5399
parch	0.2058	0.5630
pclass	0.6213	-0.2743

7. Viva Q&A

Q: What is covariance matrix?

A: C = (1/(n-1)) X^T X - matrix of all pairwise covariances.

Q: What is an eigenvalue?

A: lambda - variance along the corresponding PC direction.

Q: Why standardize before PCA?

A: So features with larger scales (e.g. fare) do not dominate.

Q: What is explained variance ratio?

A: EVR_k = lambda_k / sum(lambda_i) - fraction of total variance in PC_k.

8. Conclusion

PCA reduced 5 features to 2 components retaining 67.10% variance, simplifying visualization and analysis.